

Stat 3113: Lesson 29

2^3 Factorial Experiment Analysis of Data

Motivations



2^3 factorial experiment

- Main effects: A, B, C
- Interactions between factors:
 - Two-way interactions: AB, AC, BC
 - Three-way interaction: ABC
- Treatment combinations: there are $2^3 = 8$
 - Traditionally denoted with lowercase letters
 - The presence of a letter indicates that a factor is at its “high” level
 - “1” denotes the treatment combination when all factors are at their “low” level
 - 1, a, b, ab, c, ac, bc, abc
- Treatment combination means: $\bar{X}_1, \bar{X}_a, \bar{X}_b, \bar{X}_{ab}, \bar{X}_c, \bar{X}_{ac}, \bar{X}_{bc}, \bar{X}_{abc}$

ANOVA Table for 2^3 Factorial Experiments

- The vertical ordering differs depending on what software is used.
- We are not covering the interpretation of p_{Tr} in this course.

Source of Variation	d.f.	SS	MS	F	p-value
Treatment Combination	7	$SSTr$	$MSTr$	$F_{Tr}^* = MSTr/MSE$	p_{Tr}
A	1	SSA	MSA	$F_A^* = MSA/MSE$	p_A
B	1	SSB	MSB	$F_B^* = MSB/MSE$	p_B
C	1	SSC	MSC	$F_C^* = MSC/MSE$	p_C
AB	1	$SSAB$	$MSAB$	$F_{AB}^* = MSAB/MSE$	p_{AB}
AC	1	$SSAC$	$MSAC$	$F_{AC}^* = MSAC/MSE$	p_{AC}
BC	1	$SSBC$	$MSBC$	$F_{BC}^* = MSBC/MSE$	p_{BC}
ABC	1	$SSABC$	$MSABC$	$F_{ABC}^* = MSABC/MSE$	p_{ABC}
Error	$8(K - 1)$	SSE	MSE	-----	
Total	$N - 1$	SST	-----	-----	

Hypothesis Testing Procedure for 2^3 Factorial Experiments

- 1) Test for 3-way interaction (ABC)
 - Reject $H_{0,ABC}$ → 3-way interaction is present → STOP
 - Fail to reject $H_{0,ABC}$ → 3-way interaction absent → continue to next step
- 2) Test for 2-way interactions (AB, AC, BC). For each of these,
 - Reject H_0 → DO NOT test component main effects in that interaction
 - Fail to reject H_0 → component main effects can be tested
 - For example: AB interaction present → do not test A nor B main effects (may test C main effect as long as AC and BC interactions are both absent)
- 3) Test for main effects that are permitted by previous step

Example Setup and Data

A 2^3 factorial experiment was conducted to estimate the effects of three factors on the yield of a chemical reaction. The factors were

A: catalyst concentration (low or high)

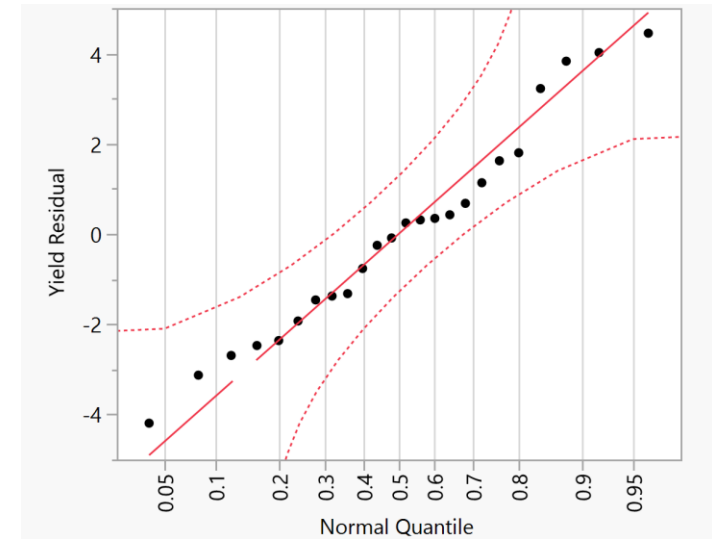
B: reagent (standard formulation or new formulation)

C: stirring rate (slow or fast).

Three replicates were obtained for each treatment. The yields, presented in the following table, are measured as a percent of a theoretical maximum.

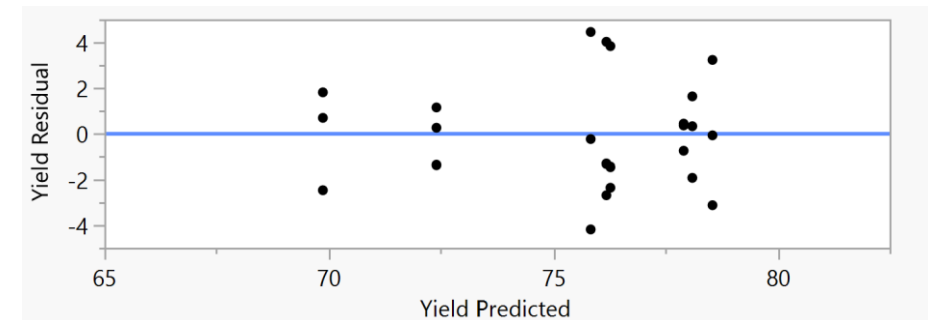
Treatment	Yield			Cell Mean
1	71.67	70.55	67.40	69.8733
<i>a</i>	78.46	75.42	81.77	78.5500
<i>b</i>	77.14	78.25	78.33	77.9067
<i>ab</i>	79.72	76.17	78.41	78.1000
<i>c</i>	72.65	71.03	73.54	72.4067
<i>ac</i>	80.10	73.91	74.81	76.2733
<i>bc</i>	80.20	73.49	74.86	76.1833
<i>abc</i>	75.58	80.28	71.64	75.8333

Normality
Assumption



Example Check conditions

Equal
Variance Assumption



Example

Use a significance level of 0.05.

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	75.640833	0.571629	132.33	<.0001*
Catalyst[low]	-1.548333	0.571629	-2.71	0.0155*
Reagent[new]	1.365	0.571629	2.39	0.0296*
Catalyst[low]*Reagent[new]	1.5875	0.571629	2.78	0.0135*
Stir rate[fast]	-0.466667	0.571629	-0.82	0.4263
Catalyst[low]*Stir rate[fast]	0.6691667	0.571629	1.17	0.2589
Reagent[new]*Stir rate[fast]	-0.530833	0.571629	-0.93	0.3669
Catalyst[low]*Reagent[new]*Stir rate[fast]	-0.533333	0.571629	-0.93	0.3647

Example Interpretation of ANOVA

Use a significance level of 0.05.

- ABC interaction is insignificant ($P\text{-value} = 0.365 > 0.05$). Then we can continue to check the two-way interaction.
- AC, BC interaction is insignificant; AB interaction is significant ($P\text{-value} = 0.013 < 0.05$). With that, we can only check the main effect of C.
- $P\text{-value}$ for factor C is 0.426, which is > 0.05 , and that means the main effect of factor C (stir rate) is insignificant.

Example Final Conclusions

- The reagent and catalyst interact to significantly impact the average chemical yield.
- The reagent and catalyst cannot be tested individually.
- Stirring rate is not significantly impacting the average chemical yield.



Reading

- CHAPTER 9 – SECTIONS 9.5

Next Topic



More Examples - JMP Analysis



Chapter 9 – Sections 9.5